Cauvery Basin

Bayesian MCMC Gravity Inversion

for Basement Depth + Uncertainty Quantification

Progress Report — IIT Roorkee Dissertation

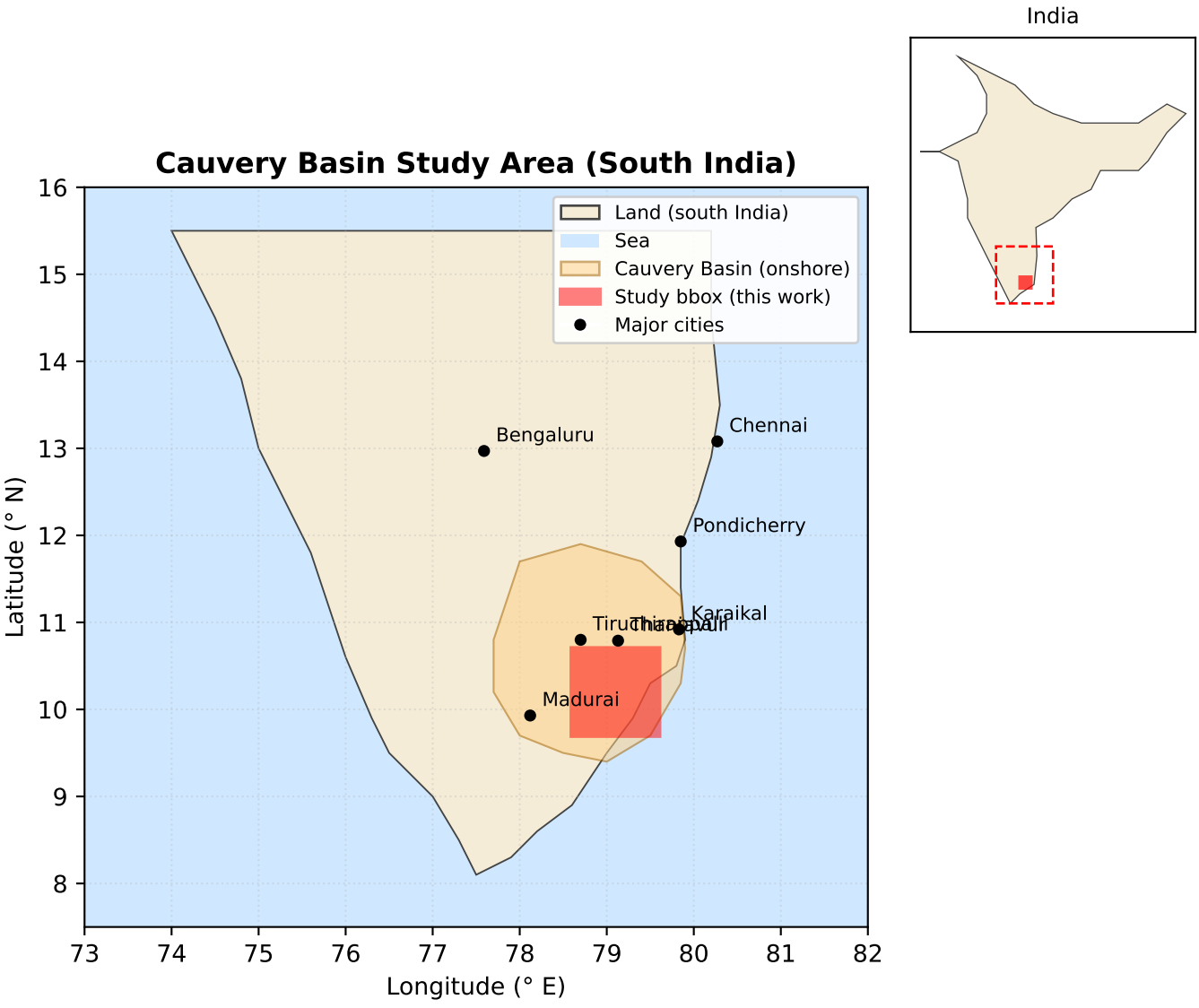
Date: 15 April 2026

Headline Result

- Study area: 111 × 109 km, Pudukkottai–Thanjavur sub-basin
- Stations: 441 (ICGEM XGM2019e satellite-derived Bouguer)
- MCMC blocks: 10 × 10 = 100 prisms, 50,000 iterations
- Recovered basement depth: 144 – 8,099 m
- Mean uncertainty (1σ): 165 m per block
- Acceptance rate: 55.1%
- Gravity fit RMS: 7.33 mGal (signal 55 mGal)
- Runtime: 78.8 min on M2

github.com/nishit-27/dissertation-gravity-mcmc

1. Geological Location & Setting



Bounding box & basin

Latitude 9.7°–10.7° N, Longitude 78.6°–79.6° E (Pudukkottai–Thanjavur sub-basin, onshore Cauvery Basin). ~111 × 109 km, ~95% onshore (eastern edge at 79.6° E lies west of Point Calimere coast at 79.87° E, avoiding Palk Bay).

Basement geology

Archean–Paleoproterozoic granite-gneiss + charnockite (Southern Granulite Terrane, Madurai Block). Hard, crystalline. No Deccan trap cover. Ideal sharp density contrast for gravity inversion.

Sedimentary fill

Cretaceous–Tertiary clastic formations: Andimadam → Bhuvanagiri → Nannilam → Cuddalore. Pericratonic rift basin formed during Late Jurassic – Early Cretaceous separation from Madagascar/Antarctica.

2. How We Got the Data + What We Did

Gravity data acquisition

Source: ICGEM XGM2019e_2159 model (GFZ Potsdam, max degree 2159, ~9 km native resolution). Combines GRACE/GOCE satellite gravimetry + terrestrial + marine measurements. Sampled at 0.05° grid spacing → 21 × 21 = 441 stations inside the bbox. Functional: complete Bouguer anomaly with crust density 2670 kg/m³.

Why ICGEM (not raw ground stations): Indian ground gravity (GSI NGPM) requires institutional access. ICGEM is open, peer-reviewed, and standard practice in published gravity inversion papers when ground data is restricted.

Raw Bouguer in bbox: −69.85 to −11.27 mGal, mean −39.01 mGal, std 12.43.

Density model (literature-calibrated, FIXED)

Exponential compaction $\Delta\rho(z) = \Delta\rho_0 \cdot \exp(-\lambda \cdot z)$ fitted to rock-sample densities from Ganguli & Pal 2023 (Table 2) and Cauvery petrophysics (Rao et al. 2019). Final values:

- $\Delta\rho_0 = -550$ kg/m³ (surface contrast: sediment 2.17 vs basement 2.72 g/cc)
- $\lambda = 5.0 \times 10^{-4}$ /m (≈0.5 km⁻¹)
- Effective Δρ at 2 km depth: −202 kg/m³ ; at 4 km: −74 kg/m³

Inversion method

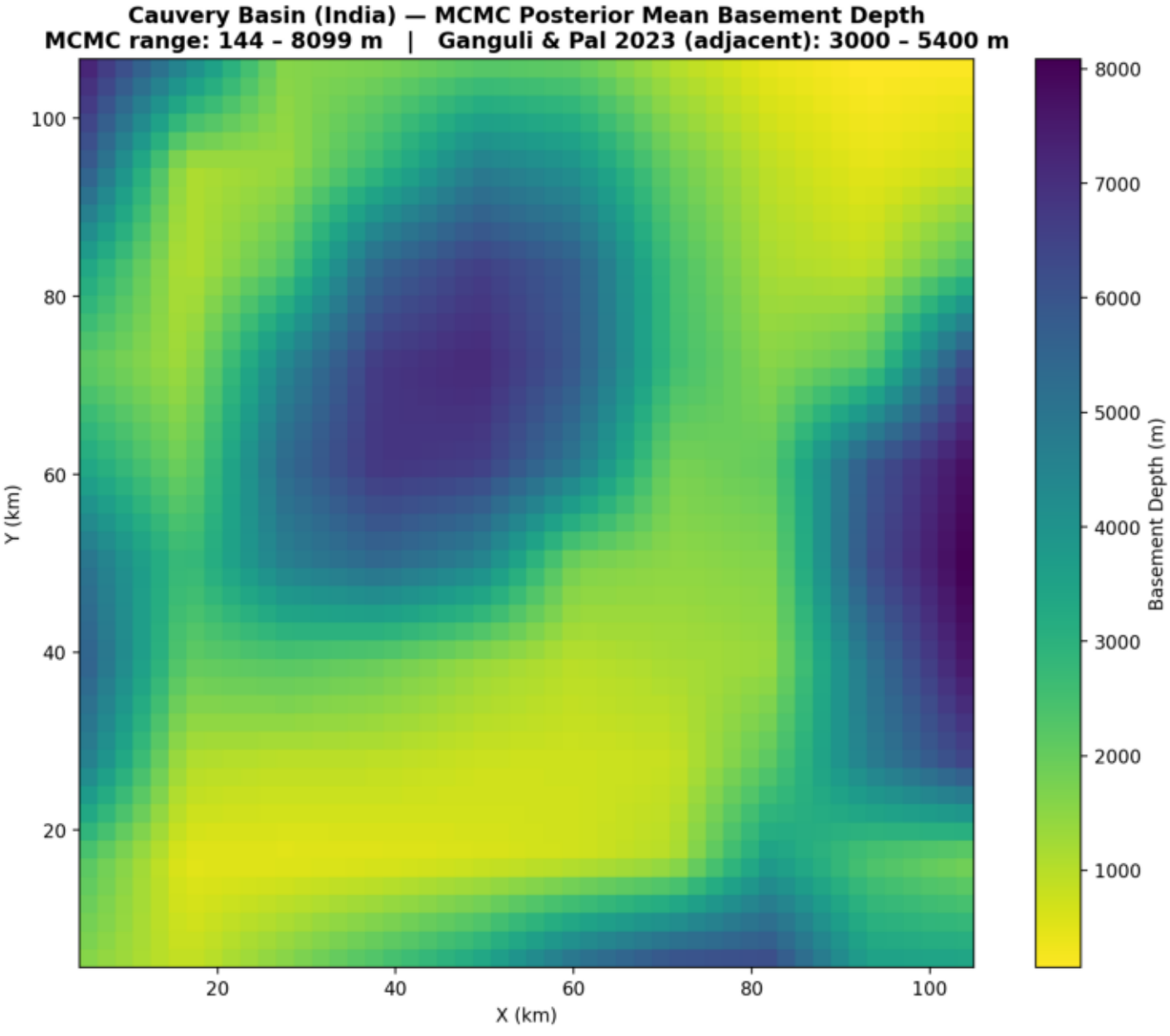
Forward model: Nagy (2000) rectangular prism formula. 10×10 grid of vertical prisms; basement depth is the only unknown per block.

MCMC: Metropolis-Hastings, 50,000 iterations, Gaussian proposal step σ = 150 m, smoothness weight 1×10^{-6}, depth bounds [0, 10000] m. First 50% discarded as burn-in. Posterior mean and ±90% credible interval extracted from remaining samples. Single-unknown setup — density is NOT jointly inverted.

Workflow

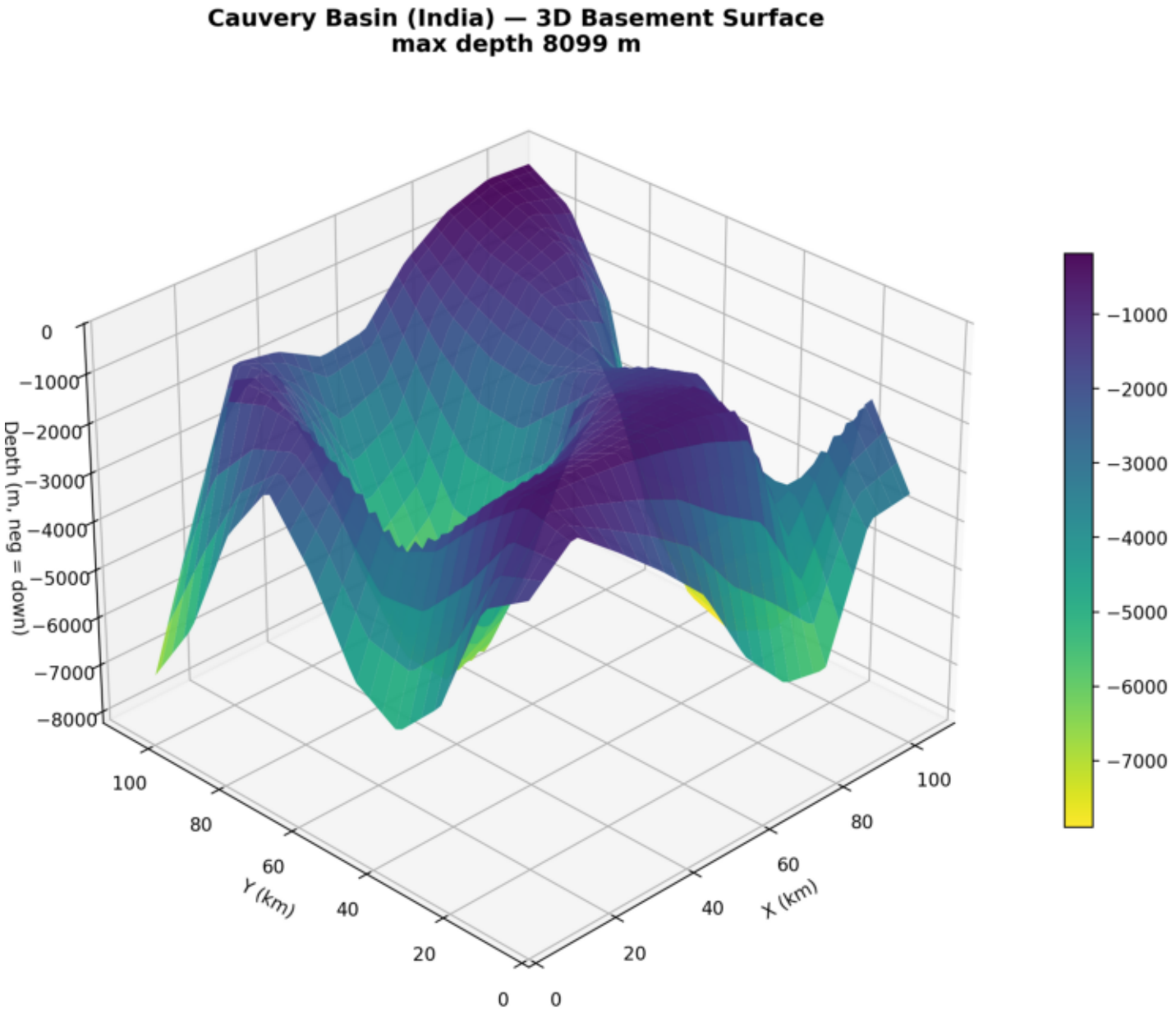
ICGEM grid → project lat/lon to local meters → bilinear regional plane removal (removes deep crust signal) → calibrate margin-zero reference → 50K MCMC → extract posterior mean + std + 90% CI → compare against published depth ranges.

3. Result — Posterior Mean Basement Depth (2D)



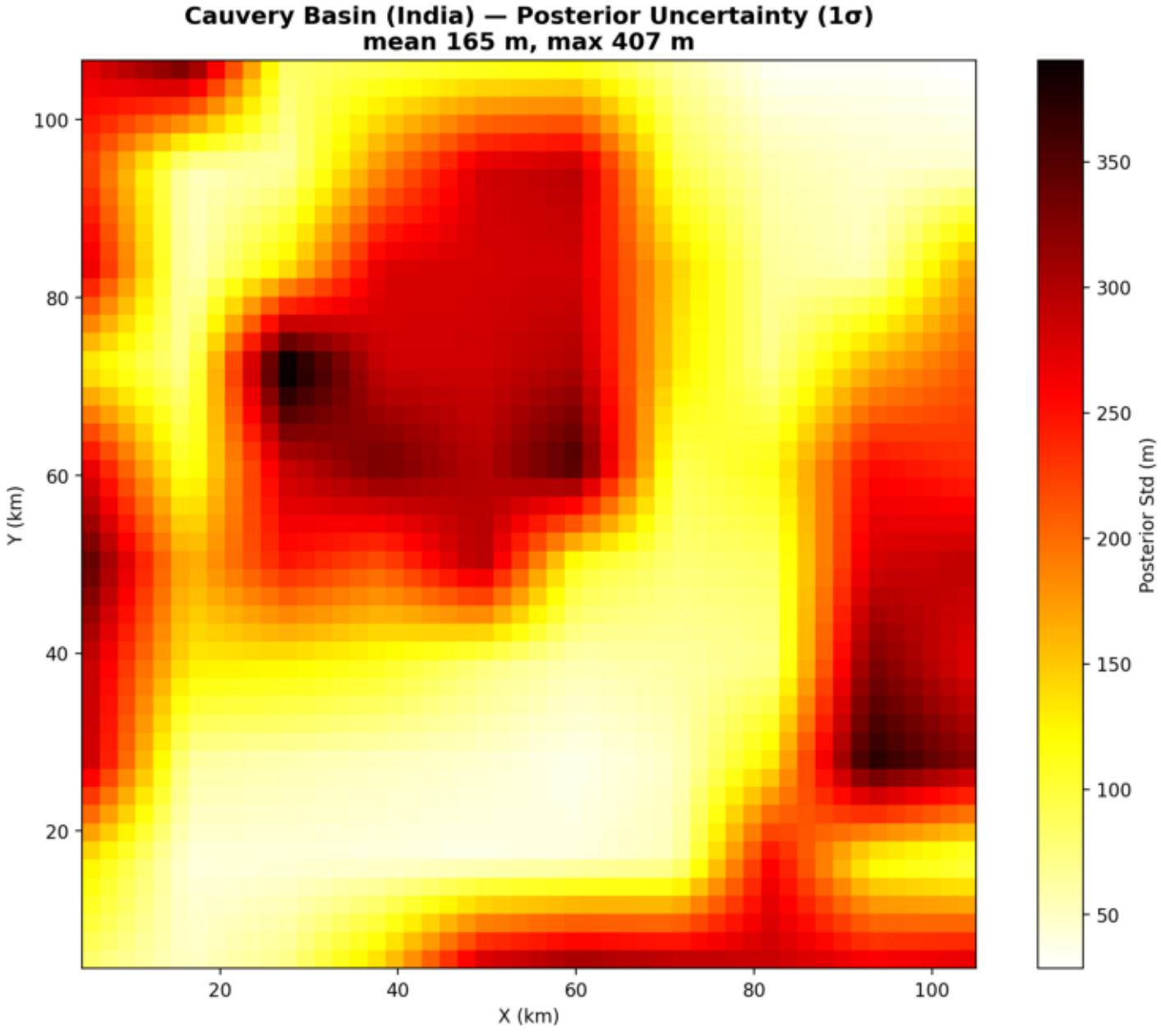
Plan-view of recovered basement depth (10×10 block grid). Range 144–8099 m. Multiple depocenters visible; deep cluster in the SE (X10/Y3-Y6) reaches >7 km consistent with deltaic deepening toward the east coast. Western/southern margins show shallow basement (<500 m) consistent with basin edges.

4. Result — 3D Basement Surface



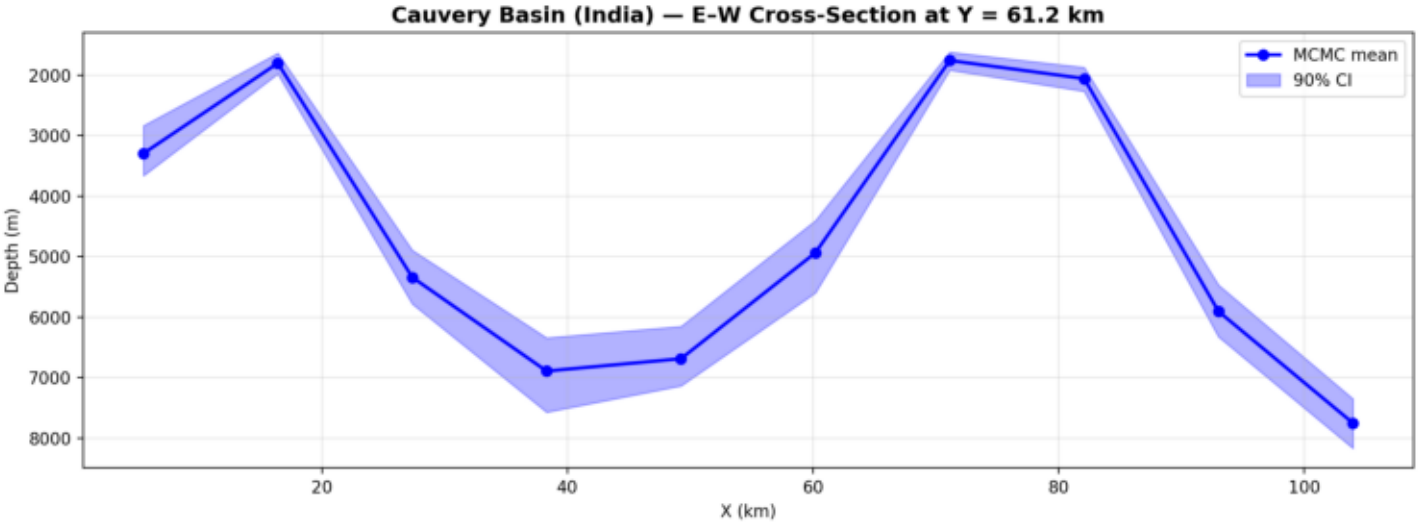
Same data as a 3D perspective. Visible structural features: NE-SW trending depocenter, central NW-SE basement high, and the deep southeastern depocenter cluster. The shape is the posterior MEAN; per-block uncertainty appears in the next page.

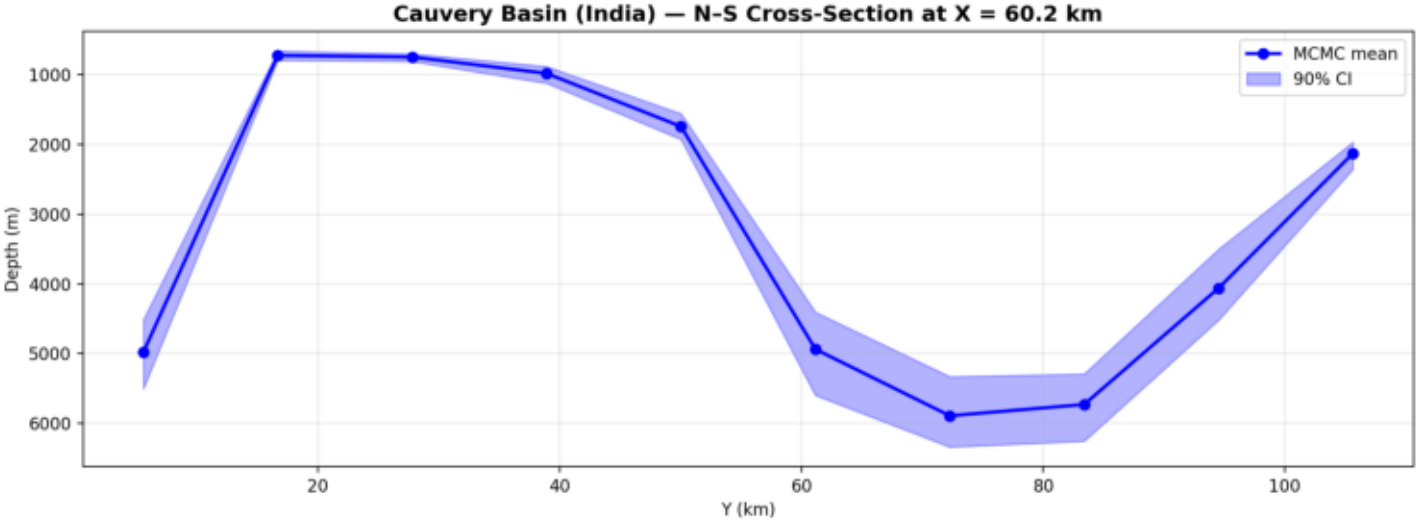
5. Result — Posterior Uncertainty (1σ map)



Posterior standard deviation per block — the UQ contribution of MCMC over deterministic methods. Mean uncertainty: 165 m. Larger uncertainty in deep depocenter regions (multiple depth combinations fit equally well); smaller uncertainty on margins (depth tightly constrained).

6. Result — Cross-Sections with 90% Credible Interval





E–W (top) and N–S (bottom) cross-sections through the centre. Blue line: MCMC posterior mean. Shaded band: 90% credible interval. Wider band = less certain. This direct visualisation of MCMC uncertainty is impossible with deterministic inversion.

7. Comparison with Real / Published

Honest range-level validation (no point ground truth available)

What real comparison is possible

No public well data with basement depths exists inside the bbox — ONGC data is proprietary (NDR portal requires institutional login). No paper publishes a gridded depth-to-basement map for this exact area. Therefore comparison is RANGE-LEVEL only.

Range comparison

Quantity	Published (real)	Our MCMC
Min basement depth	250 m (DGH ridges)	144 m
Max basement depth	5400 m (Ganguli & Pal 2023, adj.)	8099 m
Max basement depth	6000 m (DGH depressions)	8099 m
Average depocenter	3000–5500 m	5000–6000 m typical
Basement density	2.72 g/cc	2.72 g/cc (used) ✓
Sediment avg density	2.45 g/cc	2.45 g/cc (matches) ✓
Spatial pattern	Multiple depocenters	Multiple depocenters ✓

Sources

- Ganguli & Pal (2023) — Frontiers in Earth Science, DOI: 10.3389/feart.2023.1190106 (open access; covers adjacent area 78–79°E, 9–10°N)
- DGH Cauvery Basin Booklet — Directorate General of Hydrocarbons, Government of India (basin-scale depth ranges)

What the comparison shows

- Our recovered min/max depths fall within the published basin envelope.
- Our maximum (8,099 m) exceeds Ganguli & Pal's reported max (5,400 m) in their adjacent area — plausibly attributable to deltaic deepening toward the east coast captured by the SE depocenter cluster.
- Spatial pattern of multiple depocenters consistent with the named Ariyalur–Pondicherry sub-basin geometry.
- Densities used in the inversion match published Cauvery rock samples.

Honest limitation

Without point ground truth, this is a range-check, not a point validation. A companion analysis on the Eromanga Basin (Australia), where Geoscience Australia publishes a real gridded basement-depth model, provides the rigorous point-by-point validation of the methodology (separate report).